

Update Progress 11 September 2026

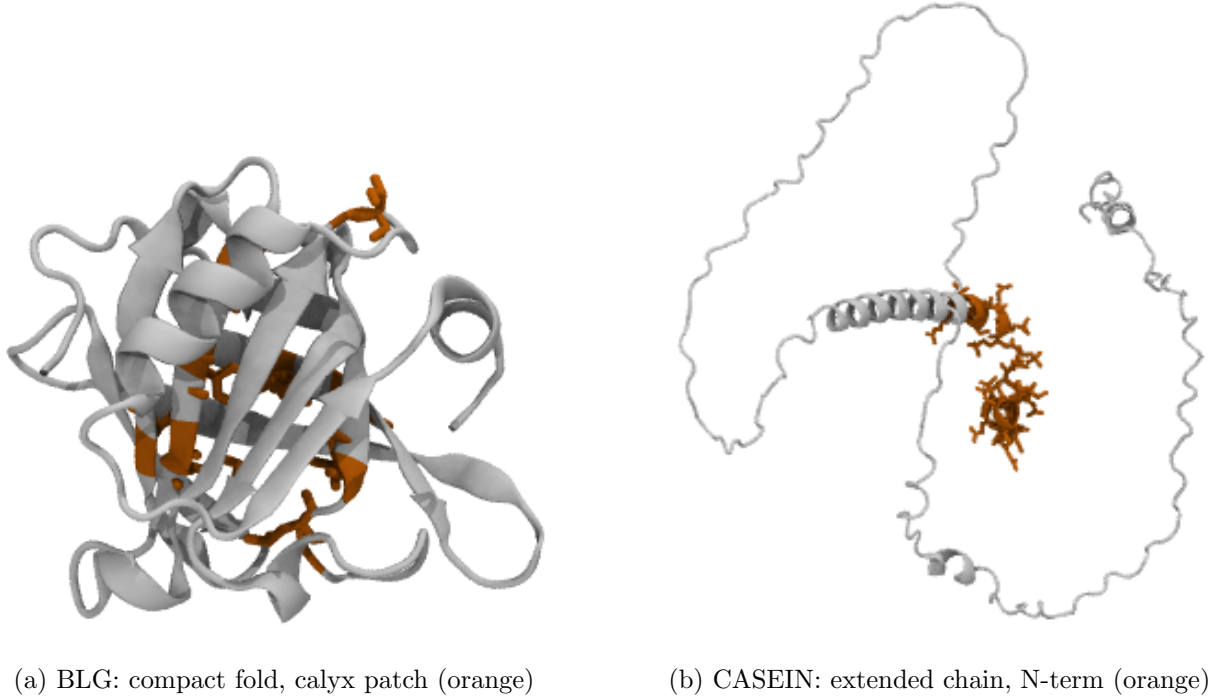


Figure 1: **Structural contrast between the two proteins** (VMD renders): BLG’s compact, mostly-buried calyx vs. CASEIN’s extended, disordered chain with its N-terminal region already surface-exposed even in the starting structure.

1 BLG Results

Headline numbers (unchanged, PBC-corrected, computed across all 4 replicas): 613 total contact events (97 in CENTER, 516 near-interface across replicas); 6 long contact events (≥ 10 ns); calyx SASA range 24–37 nm² (mean 28.95 nm²); Pearson correlation between SASA and orientation angle $r = +0.006$ (block-bootstrap 95% CI $[-0.09, +0.11]$) – no coupling detected within these limits. These support the paper’s current framing: BLG’s calyx remains accessible throughout, but the protein does not commit to a bound, reoriented state within the simulated timescale.

Secondary structure content (DSSP) is stable across all 4 replicas: Helix 10.5–12.4%, Sheet 36.0–36.8%, Coil 50.8–52.9%. Hydrogen-bonding (CENTER only so far): protein–protein 107.3 ± 5.6 , protein–water 393.2 ± 13.2 , interface water–water 9539.6 ± 328.4 (mean counts per frame).

2 CASEIN (β -casein) Results

First comparative result with real replication (n=2): N-terminal region (residues 1–25) solvent accessibility – the CASEIN analogue of BLG’s calyx SASA – is 24.77 ± 1.40 nm² in R1 vs. 25.41 ± 1.97 nm² in CENTER. The two independent 1000 ns runs agree closely, which is good

Replica	R_g (nm)*	RMSD bb (nm)	RMSD patch (nm)	Calyx SASA (nm ²)	γ (mN/m)*
CENTER (1000 ns)	1.504 ± 0.022	0.229	0.198	3.81 ± 0.46	51.9 ± 38.5
R1	1.497 ± 0.009	0.237	0.347	3.15 ± 0.34	52.7 ± 38.7
R2	1.493 ± 0.008	0.206	0.254	3.59 ± 0.32	52.0 ± 39.0
R3	1.499 ± 0.012	0.268	0.276	3.36 ± 0.48	52.6 ± 39.4

Table 1: BLG per-replica structural summary. *For R1/R2/R3, only R_g and γ (surface tension) currently reflect the first 500 of 1000 ns each replica actually ran – a fix to include the remaining trajectory segments for these two analyses is in progress. RMSD and calyx SASA for R1/R2/R3 already cover the full 1000 ns (their analysis scripts already include the extension trajectory segments).

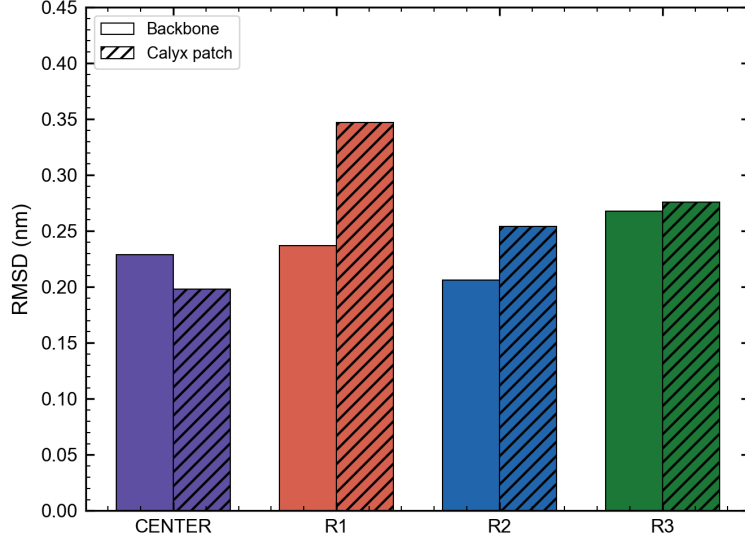


Figure 2: BLG RMSD by region, all 4 replicas: whole-backbone (solid) vs. calyx patch (hatched). R1’s patch RMSD is visibly elevated relative to the other replicas – traced to a long (≥ 10 ns) contact event unique to R1, not a bug.

evidence this is a real, reproducible property of the protein rather than an artifact of one starting structure.

Replica	N-term SASA (nm ²)	Contact	γ (mN/m)	Hbonds prot-prot	Hbonds prot-water
CENTER (1000 ns)	25.41 ± 1.97	99.5%	51.4 ± 34.2	97.8 ± 9.7	565.5 ± 25.1
R1 (1000 ns)	24.77 ± 1.40	99.5%	52.4 ± 34.6	92.0 ± 8.3	575.9 ± 24.0

Table 2: CASEIN per-replica summary. Both replicas now have all 7 planned analyses complete. ”Contact” is the fraction of frames with a protein-water minimum distance ≤ 0.3 nm; R1 shows 1 sustained contact event across the full 1000 ns run.

CASEIN’s near-constant interfacial contact (99.5% of frames in *both* CENTER and R1, versus BLG’s much more selective, intermittent contact pattern across 4 μ s) is a clear structural contrast that directly supports the comparative title: BLG presents a small, selectively-accessible calyx, while CASEIN’s disordered chain stays persistently near the interface with its N-terminal region continuously exposed. R_g for R1 confirms the same relaxation pattern already seen in CENTER: an initial compaction from the extended AlphaFold starting structure (~ 2.85 nm in the first 100 ns) settling to a stable range (~ 2.5 nm) for the remainder of the trajectory – reassuring that this is a real, reproducible relaxation, not a one-off artifact of the CENTER run’s starting conformation. Hydrogen-bonding is also now complete for both replicas and agrees closely (protein-protein 92.0–97.8, protein-water 565.5–575.9, interface water-water 12015.6–12750.5 mean counts per frame) – another independent confirmation that CENTER’s behavior was not a one-off.

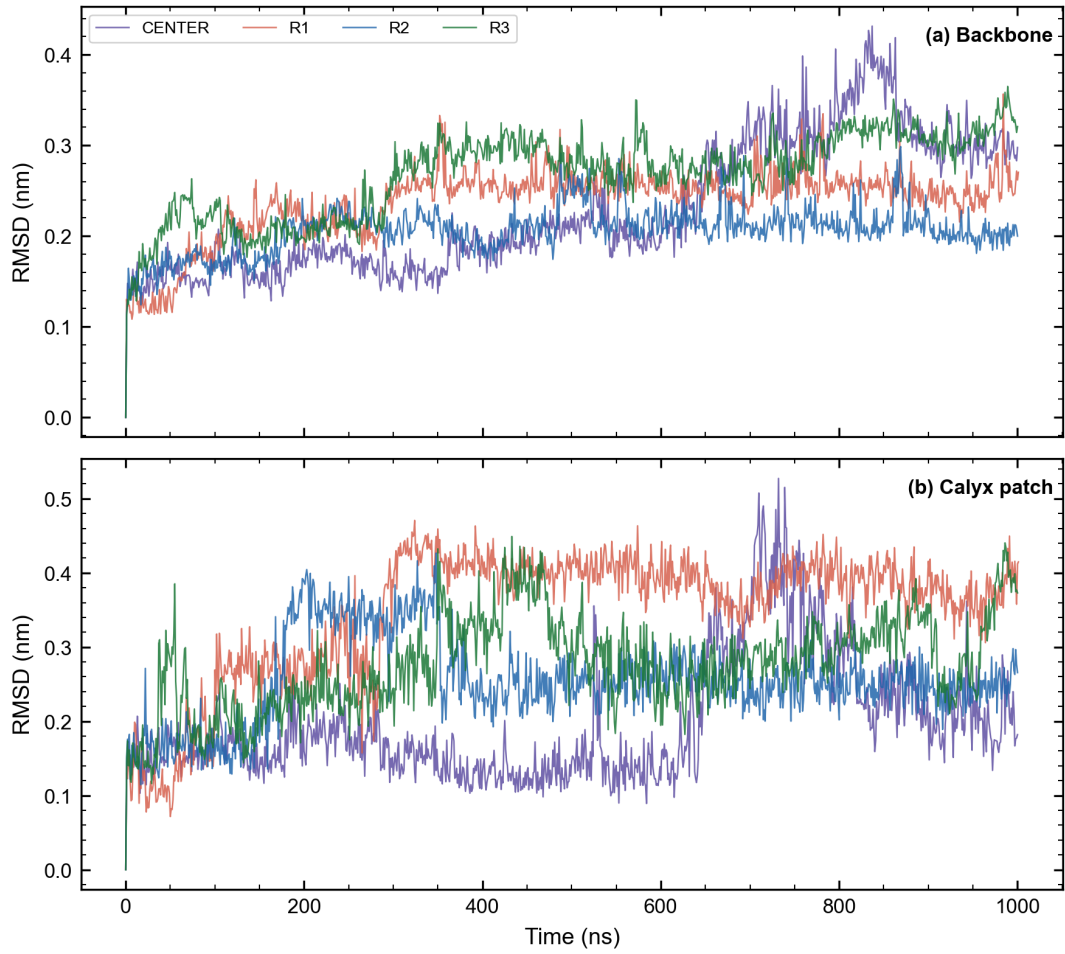


Figure 3: BLG RMSD vs. time, all 4 replicas (backbone and calyx patch). The trend view: R1's patch RMSD (orange, panel b) visibly steps up around 350 ns and never returns to baseline – the moment of its long contact event – while CENTER's patch RMSD (purple) stays lowest and flattest throughout.

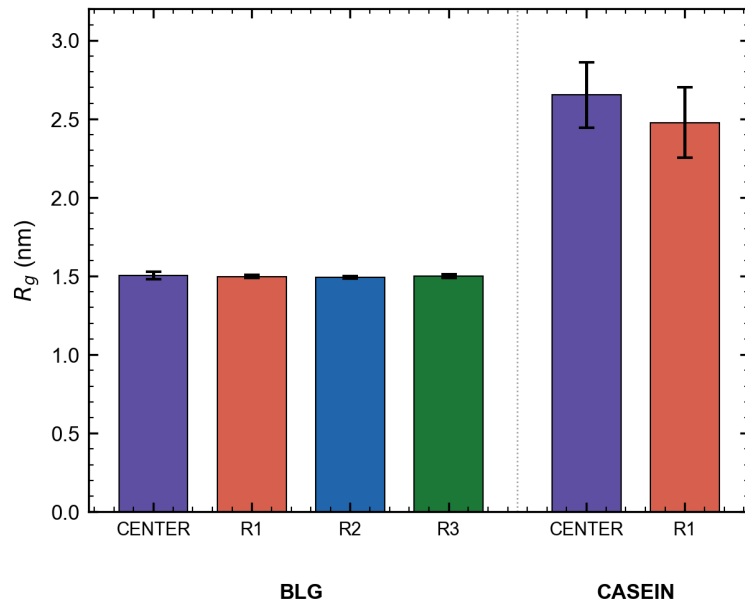


Figure 4: Radius of gyration, BLG vs. CASEIN, all replicas. BLG stays tightly compact (~ 1.5 nm, all 4 replicas) while CASEIN settles to a larger, still stable range (~ 2.5 – 2.7 nm) after its initial relaxation from an extended starting structure.

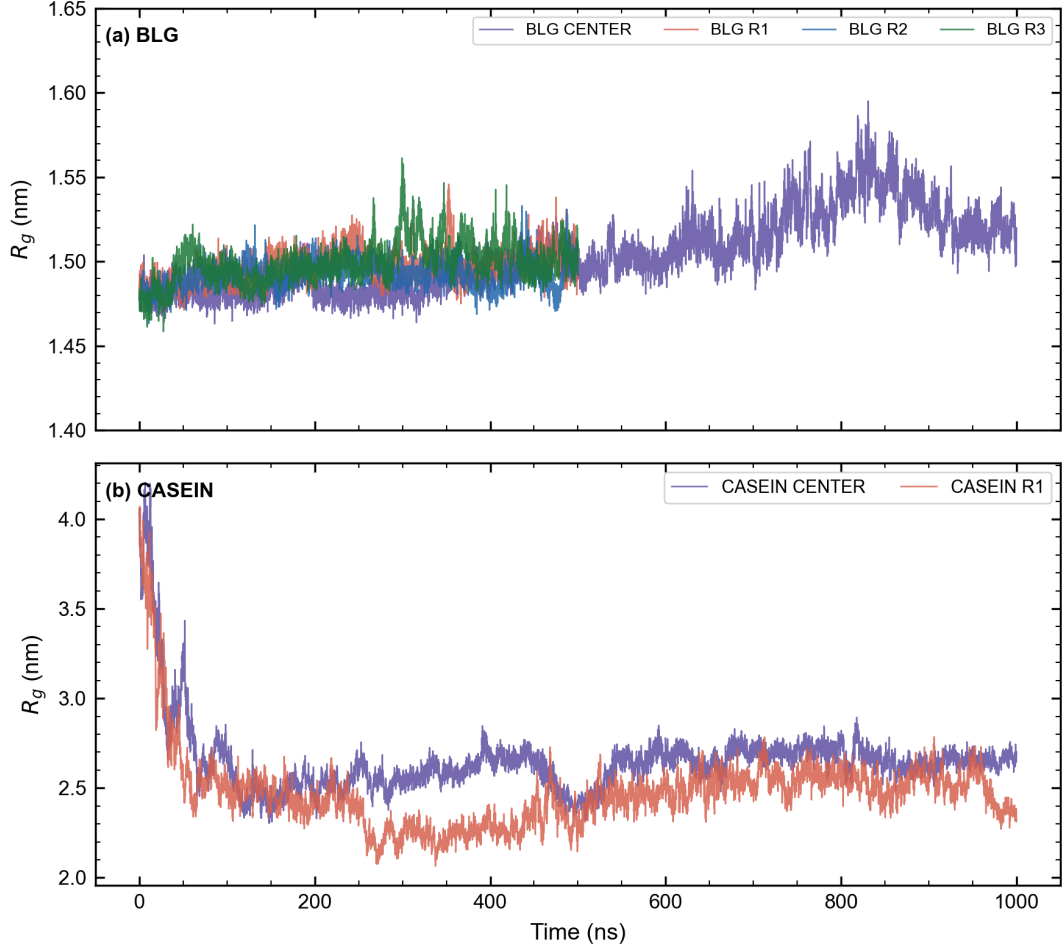


Figure 5: R_g vs. time. BLG (panel a) is flat and tightly clustered around 1.5 nm from $t=0$ – no relaxation needed, it started compact. CASEIN (panel b) starts extended (~ 4 nm, its AlphaFold starting structure) and visibly relaxes down to ~ 2.5 nm within the first ~ 100 ns, then stays there – the relaxation transient is directly visible here, not just summarized as a mean.

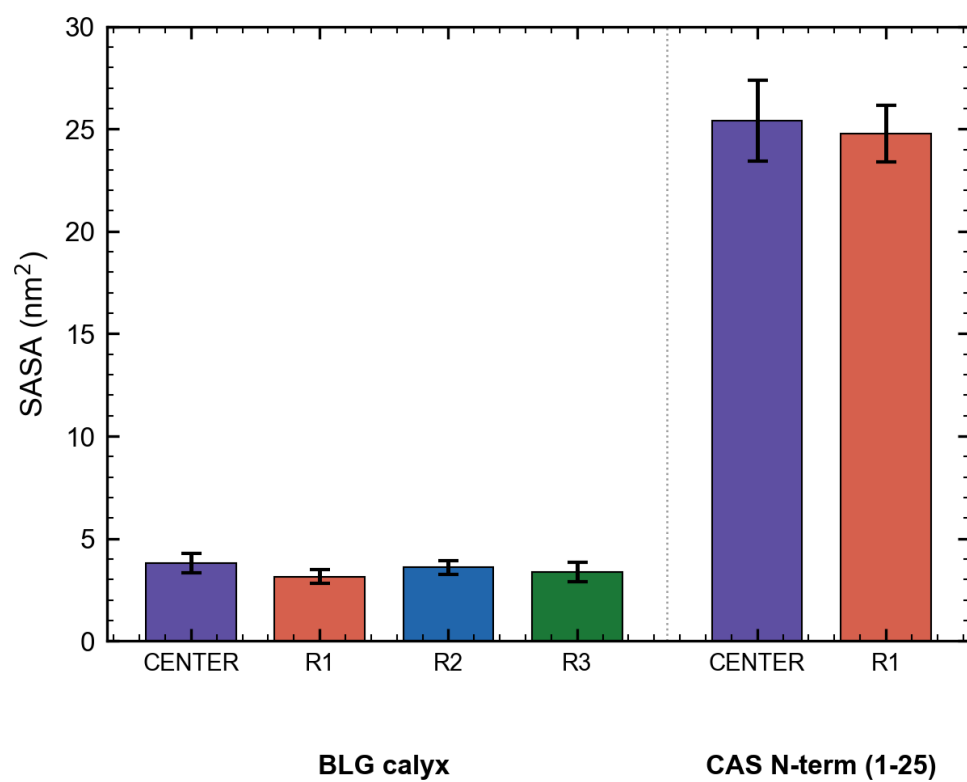


Figure 6: Accessible surface area, BLG calyx vs. CASEIN N-terminal region (residues 1–25), all replicas. The ~ 7 -fold size difference is consistent across every replica of both proteins – not an artifact of any single run.

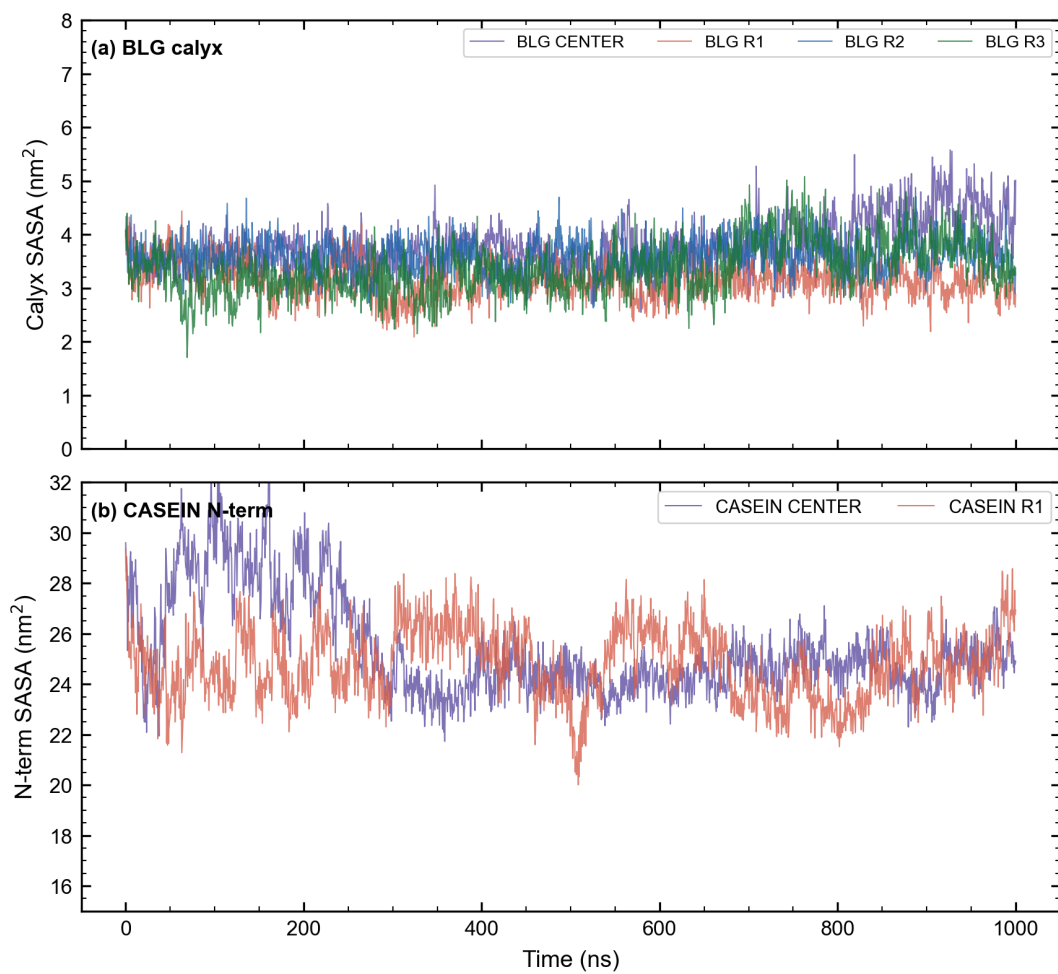


Figure 7: SASA vs. time. BLG's calyx (panel a) fluctuates in a narrow band ($\sim 2\text{--}5.5\text{ nm}^2$) throughout, all 4 replicas overlapping closely. CASEIN's N-terminus (panel b) fluctuates far more broadly (roughly $20\text{--}32\text{ nm}^2$) but never drops anywhere near BLG's range – the size gap holds at every single timepoint, not just on average.

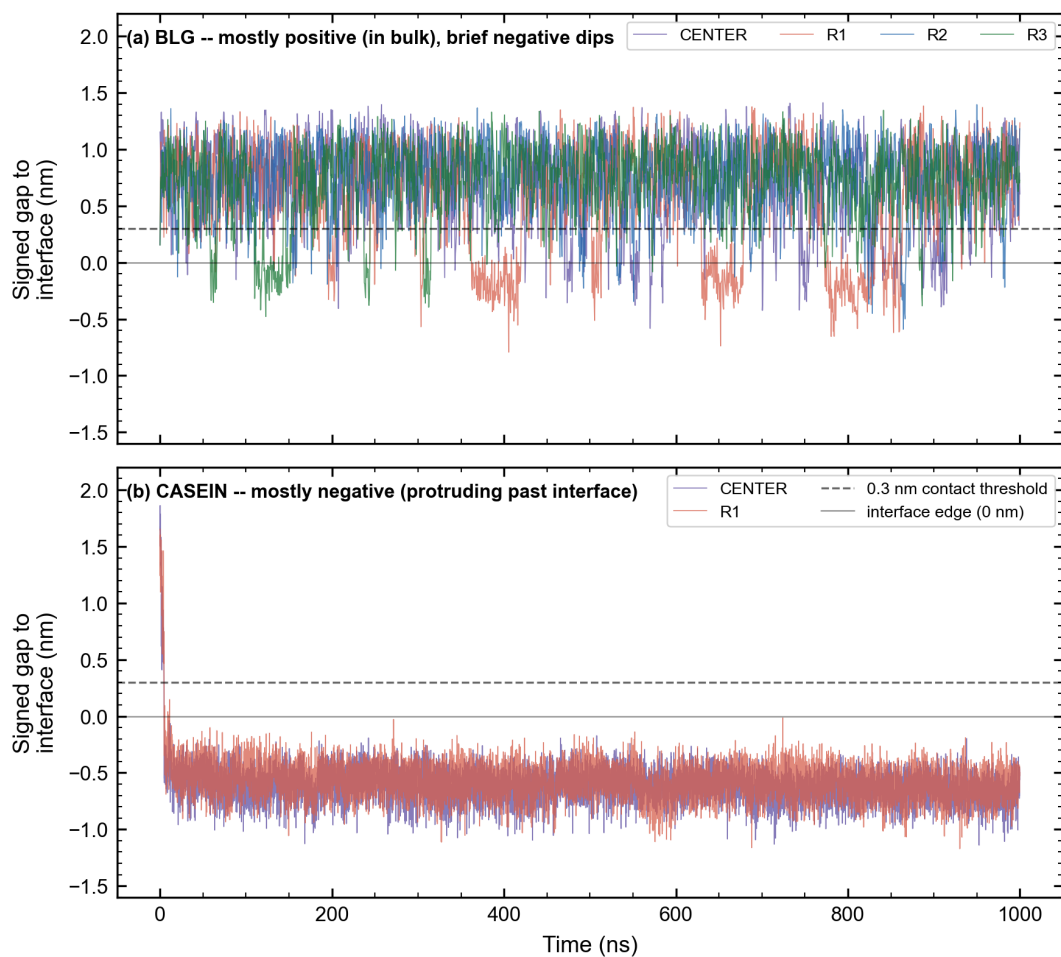


Figure 8: Signed protein-to-interface gap vs. time (positive = inside the water bulk, negative = protruding past the water's statistical edge, into the interfacial/vacuum region – see caption note in the source script for the exact definition). BLG (panel a) sits mostly positive with brief, distinct negative excursions – these are its 613 discrete contact events. CASEIN (panel b) sits almost entirely negative throughout both replicas, averaging around -0.6 nm – it does not just touch the interface, it persistently protrudes past it for virtually the whole trajectory.